Native finite-size extensions: implementation receipt

Bounded development correctness; no performance claim

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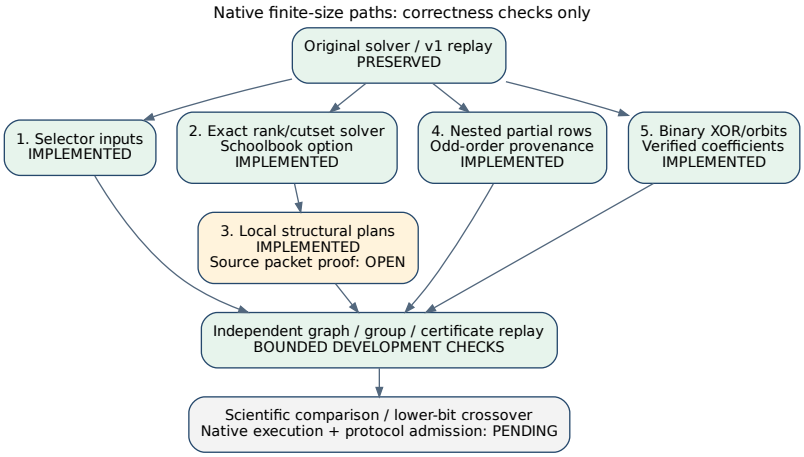
The user requested implementation of review items 1–5. These add opt-in selector, exact-contraction, partial-relation and binary/orbit paths. The original solver, seeded defaults, certificate replay and source packet guards remain available. No scientific experiment or benchmark ran.



1. Selector inputs: implemented and development-checked. The engine branches on inverse-hash interval values, caches bounded decodes, retains slot identities and optionally derives the final point from the sum equation. A separate tensor encoder makes selectors the actual free inputs. All returned points pass independent subgroup, hash and sum replay. Decoder randomness is separate from the label RNG. The alternative graph counts decoded tuples, not cofactor-projection preimages. Generic tuple search and decoder work remain; no exponent improvement is established.

2. Encoding and solver choice: implemented and checked. A greedy rank planner and exact BigUint variable elimination admit low-rank tables and exhaustive cutset children under explicit resource policies. Witness-only search propagates exact local support without requesting count precision and checks the full graph before returning. A schoolbook/ripple product is an optional alternative to the unchanged Fourier default. Opaque predicates are not assumed cheaply factorable. No optimal-width guarantee, SAT/CNF-XOR exporter, approximate-zero fallback or performance result is claimed.

3. Checked local plans: partial against the proposal. Admission recomputes the graph/pin edge partition and all intermediate ranks; execution checks live storage and covers all cutset assignments. This is finite exact conditioning, not a new small-parameter Appendix D packet schedule. Smaller-packet packing, coefficient coverage, precision, reconstruction, shrinkage and cost obligations remain open. The original massive packet-recursion onset is unchanged. No new theorem or Lean proof is claimed, and bounded controls do not replace the independent audit.

4. Partial relations: implemented and checked. Nested bases require the same prime/hash seed with ordered thresholds. The bounded collector admits up to two outer points, pivots outer columns before inner columns, and retains every original labelled row and its provenance. Arithmetic is modulo the odd subgroup prime, not XOR cycle cancellation. Independent replay checks memberships, group equations, all column cancellations and scalar extraction. A known r=7 control uses coefficient 6 rather than binary coefficient 1. A separate bounded query driver generates uniform labels without exposing them to the witness solver.

5. Binary specialization: implemented components. Native polynomial arithmetic verifies irreducibility and inverses. The odd-degree Koblitz constructor checks primality proofs, generator and full cardinality by Hasse uniqueness or exact bounded

counting. Half-trace decompression is canonical. Full-rank affine XOR hashes have checked inverse fibers. Frobenius eigenvalue candidates come from the known Koblitz polynomial and are checked on the generator; representative coefficients receive scalar replay. Orbit domains expand all selected orbits, including shorter ones. Changed hash distributions need their own coverage analysis. This does not establish full official binary setup, a challenge solve or a prime-field specialization.

Validation scope, identities and remaining work

The known-answer prime fixture is p=5, a=2, b=1, r=7, h=1, G=(0,1); library target [3]G=(3,3), existing CLI target [4]G=(3,2). A second fixture has p=5, a=0, b=1, r=3, h=2, G=(0,1). Binary fixtures use $f=x^5+x^2+1$, r=11 and $y^2+xy=x^3+ax^2+1$: a=0, h=4, G=(4,22), Frobenius eigenvalue 4; a=1, h=2, G=(10,18), eigenvalue 5. The official degree-131 polynomial $x^{131}+x^{13}+x^2+x+1$ passes irreducibility checking; that is field-only validation. Complete official order/generator/target validation is not claimed.

Controls compare exhaustive small domains/targets and graph truth tables, pins and conditioned counts; forged witnesses/plans, invalid codes, singular hashes, isolated false predicates and resource exhaustion; full known-answer IC and partial recovery; original CLI replay and exclusive creation; independent small carryless products, group/cardinality/Frobenius identities, complete orbit enumeration and coefficient replay. A 70-edge witness succeeds under a one-bit precision policy while exact counting is correctly refused. The source-compatible default certificate controls stay in the full suites. Raw final transcripts, source manifests and exact commands are in README.md and VALIDATION.md. Local test durations are not benchmarks.

The parent report retains primary sources and negative SAT/enumeration evidence. Sources include Bochum v0.2, TensorOrder, WDSat, and the pinned lead atlas. No comprehensive novelty or independent mathematical audit is completed.

The original frozen A/B panel and both challenge scalars remain unresolved. Earlier S3 upload/readback succeeded; authenticated native image/execution and measurement admission remain unverified. The measurement card stays draft pending its driver, exact protocol/identities, checker, cost envelope and execution-host checkpoint. No chat credentials were reused. No performance, scaling or sub-rho conclusion follows from this receipt. Canonical graphs receive no measured value or curve-transport edge.